

"""Q1)A light bulb can be either Working (1) or Not Working (0).

Write a Python program to:

Generate 10 Bernoulli random values.

Print the generated values.

Count how many bulbs are working."""

#ANS:

```
import numpy as np
```

```
bulbs = np.random.binomial(1, 0.5, 10)
```

```
print("Generated Values:")
```

```
print(bulbs)
```

```
working = np.sum(bulbs)
```

```
print("Working Bulbs:", working)
```

Generated Values:

```
[0 0 0 1 1 1 0 0 0 1]
```

Working Bulbs: 4

"""Question 2

A coin is tossed 5 times.

Write a Python program to:

Simulate the experiment using the Binomial distribution.

Print the number of heads obtained.

Repeat the experiment 5 times."""

#ANS:

```
import numpy as np
```

```
print("Number of Heads in 5 Experiments:")
```

```
for i in range(5):
```

```
    heads = np.random.binomial(5, 0.5)
```

```
    print("Experiment", i + 1, ":", heads)
```

Number of Heads in 5 Experiments:

Experiment 1 : 3

Experiment 2 : 3

Experiment 3 : 2

Experiment 4 : 3

Experiment 5 : 2

"""Question 3

A player has a 50% chance of winning a game.

Write a Python program to:

Generate 10 Geometric random values.

Print the generated values.

Find the maximum number of attempts required for the first win."""

#ANS:

```
import numpy as np
```

```

attempts = np.random.geometric(0.5, 10)

print("Generated Values:")
print(attempts)

print("Maximum Attempts:", np.max(attempts))

```

```

Generated Values:
[1 2 3 1 2 3 2 3 1 5]
Maximum Attempts: 5

```

"""Question 4

A shop receives an average of 2 customers per hour.
Write a Python program to:
Generate customer arrivals for 10 hours.
Print the generated values.
Find the highest number of customers received in an hour."""

```

#ANS:
import numpy as np

customers = np.random.poisson(2, 10)

print("Customers per Hour:")
print(customers)

print("Highest Customers in an Hour:", np.max(customers))

```

```

Customers per Hour:
[2 1 1 2 0 3 1 0 3 0]
Highest Customers in an Hour: 3

```

"""Question 5

Write a Python program to generate 5 random values from each of the following distributions:

Bernoulli Distribution
Binomial Distribution
Geometric Distribution
Poisson Distribution

Print the generated values for each distribution separately."""

```

#ANS:
import numpy as np

bernoulli = np.random.binomial(1, 0.5, 5)

binomial = np.random.binomial(5, 0.5, 5)

geometric = np.random.geometric(0.5, 5)

poisson = np.random.poisson(2, 5)

print("Bernoulli Distribution:")
print(bernoulli)

print("\nBinomial Distribution:")
print(binomial)

```

```
print("\nGeometric Distribution:")
print(geometric)

print("\nPoisson Distribution:")
print(poisson)
```

Bernoulli Distribution:
[0 0 0 0 1]

Binomial Distribution:
[3 3 2 1 1]

Geometric Distribution:
[2 3 1 1 1]

Poisson Distribution:
[2 2 1 2 2]